

CAT 2025

MBA FASTRACK BATCH

Lecture - 03

Logical DI

Venn Diagrams 02

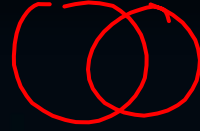
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RECAP OF THE LAST LECTURE



A → $\begin{matrix} \times & \checkmark \end{matrix}$ → 2
B → $\begin{matrix} \times & \checkmark \end{matrix}$ → 2
C → $\begin{matrix} \times & \checkmark \end{matrix}$ → 2
D → $\begin{matrix} \times & \checkmark \end{matrix}$ → ?



2 set →

4 regions

$$2 \times 2 = 4 = 2^2$$

3 set →

8

$$2 \times 2 \times 2 = 8 = 2^3$$

4-set →

→

$$2 \times 2 \times 2 \times 2 = 2^4 = 16$$

$$n\text{-sets} \rightarrow 2^n$$

TOPICS

to be covered

1

Basics of 4-set Venn diagrams & Maxima-Minima

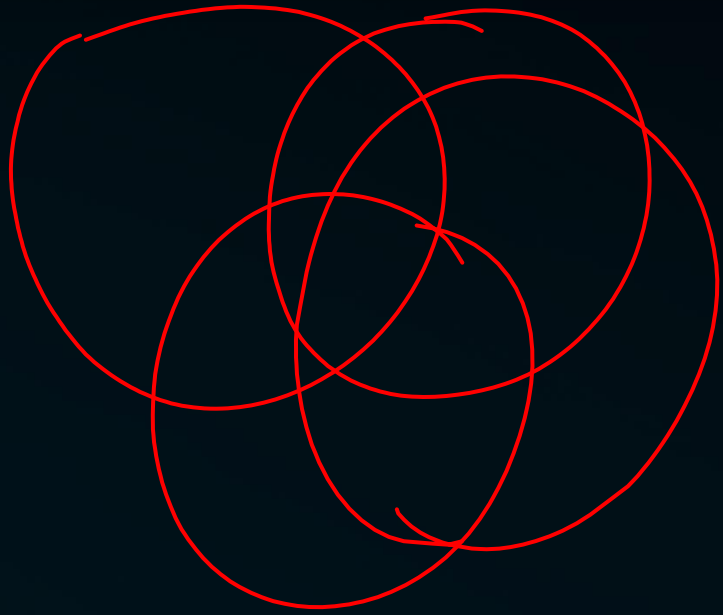
2

Questions on 4-set Venn diagrams & Maxima-Minima

2 sets

1 set

Venn Diagram : Basics



Venn Diagram : Basics



Sets $\rightarrow P, Q, R, S$

Ex₀ None - (1) ✓

Ex₁ $\rightarrow$ (4)
only P $\rightarrow a$

only Q $\rightarrow b$

only R $\rightarrow c$

only S $\rightarrow d$

Ex₂ $\rightarrow$ (6)

Only (P & Q) $\rightarrow e$

" (P & R) $\rightarrow f$

" (P & S) $\rightarrow g$

" (Q & R) $\rightarrow h$

" (Q & S) $\rightarrow i$

" (R & S) $\rightarrow j$

Ex₃ $\rightarrow$ (4)

only (P, Q & R) $\rightarrow k$

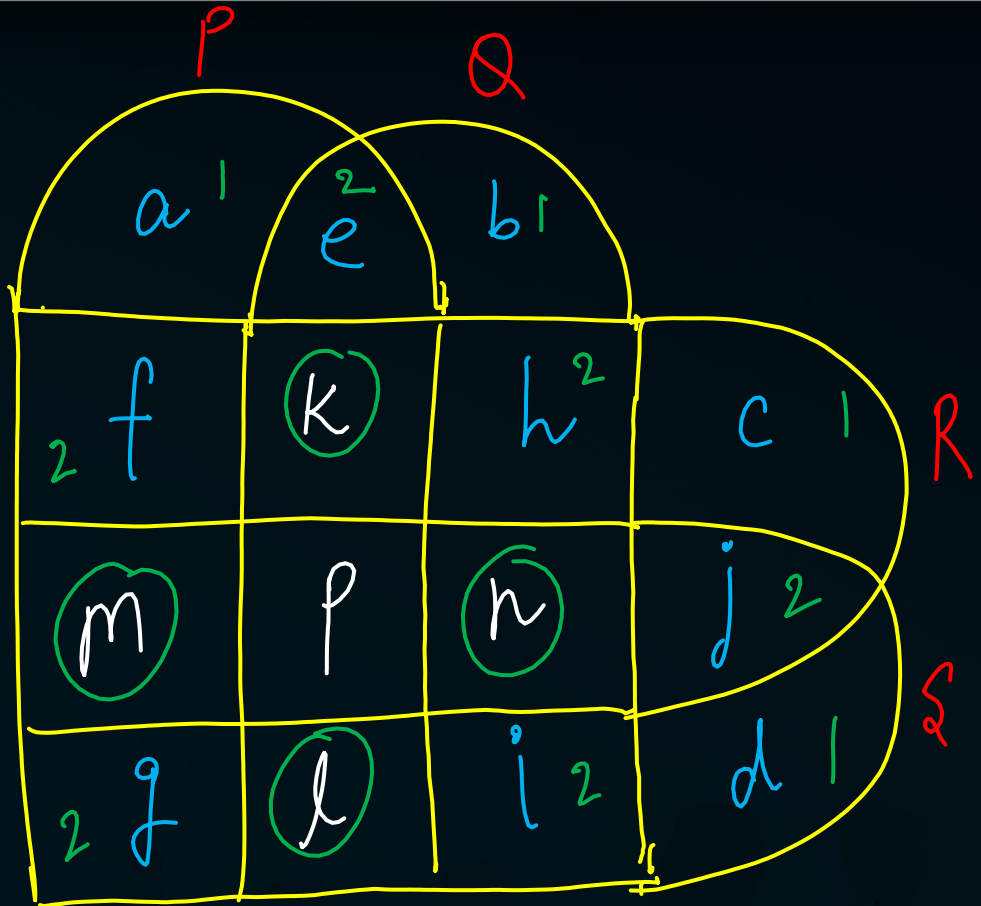
" (P, Q & S) $\rightarrow l$

" (P, R & S) $\rightarrow m$

" (Q, R & S) $\rightarrow n$

Ex₄ $\rightarrow$ (1) $\rightarrow p$

Venn Diagram : Basics



9



Set-1

Read the following data and answer the given questions.



#Q. In a group of people, each person watches at least one of the four channels, Zee TV, Colours, Sony TV, and Imagine TV. 440 watch Zee TV. 440 watch Colours, 425 watch Sony TV, and 395 watch Imagine. 50 people watch each combination of exactly three channels. 60 people watch only Zee TV and Sony TV. 90 people watch only Zee TV. 110 people watch only Colours, 100 people watch only Sony TV. 80 people watch only Imagine TV. 40 people watch only Sony TV and Imagine. 20 people watch all the four channels.

Questions



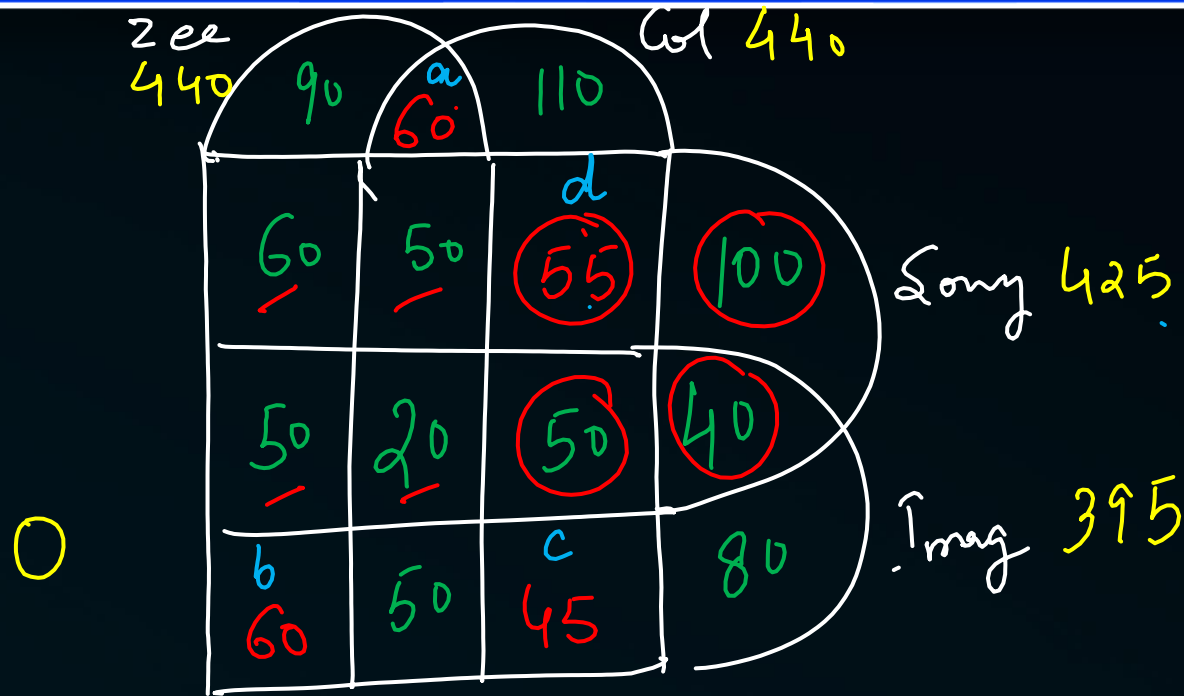
#Q. How many people watch Zee TV and Colours only?
a. 40 ☒ b. 60 c. 80 d. 100

#Q. How many people watch Sony TV but do not watch Zee TV?
a. 240 b. 235 c. 295 ☒ d. 245

#Q. How many people watch exactly two channels?
☒ a. 320 b. 360 c. 280 d. 260

#Q. How many people took part in the survey?
a. 800 b. 820 c. 850 ☒ d. 920

Analysis



$$d = 55$$

$$\begin{cases} a + b = 120 \\ b + c = 105 \\ c + a = 105 \end{cases}$$

$$a = 60$$

$$b = 60$$

$$c = 45$$

$$2(a + b + c) = 330$$

$$a + b + c = 165$$

In a group of people, each person watches at least one of the four channels, Zee TV, Colours, Sony TV, and Imagine TV. 440 watch Zee TV. 440 watch Colours, 425 watch Sony TV, and 395 watch Imagine. 50 people watch each combination of exactly three channels. 60 people watch only Zee TV and Sony TV. 90 people watch only Zee TV. 110 people watch only Colours, 100 people watch only Sony TV. 80 people watch only Imagine TV. 40 people watch only Sony TV and Imagine. 20 people watch all the four channels.



Max-Min

Venn Diagram : Basics

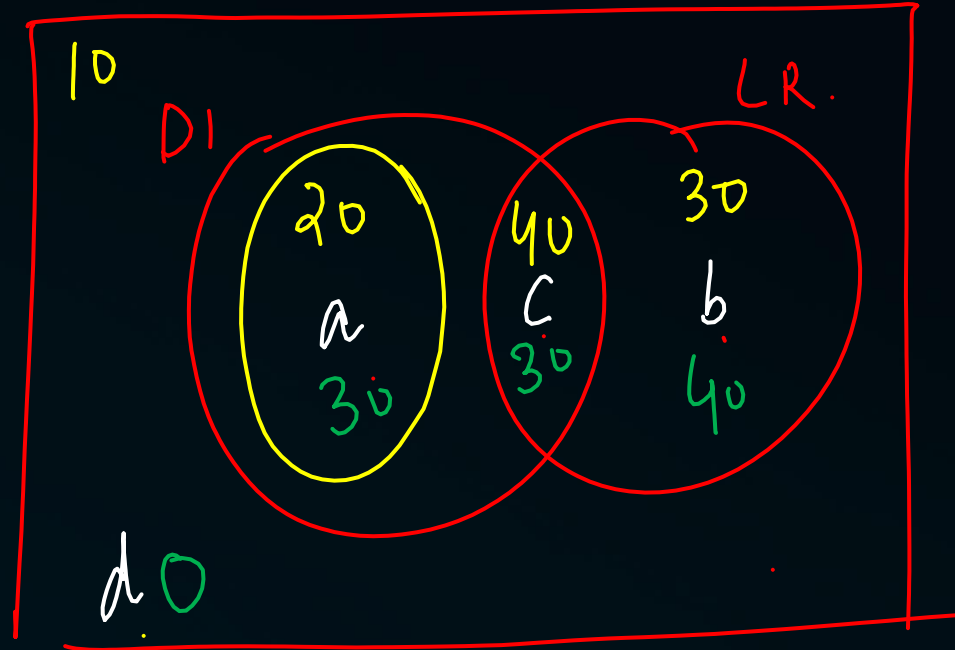


100 → Total.

60 → DI

70 → LR

None → 0
→ 10



$$a + b + c + \underline{d} = 100$$

$$a + c = 60$$

$$b + c = 70$$

$$a + b + c = 90$$



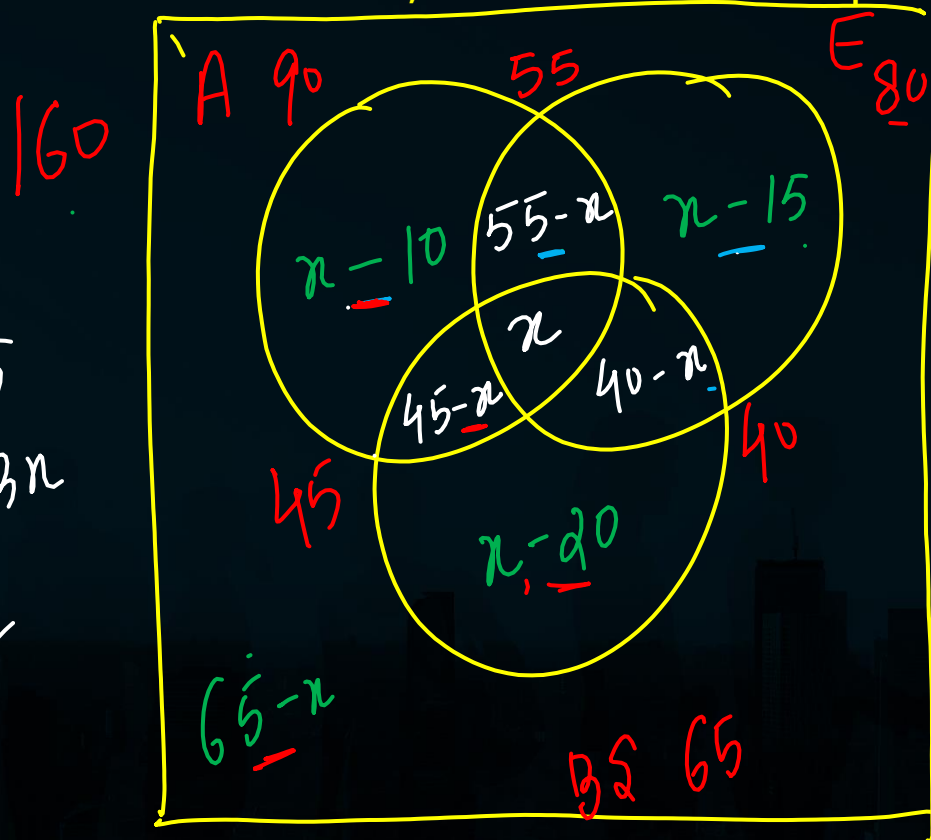
Set-2

Read the following data and answer the given questions.



#Q. Among the 160 students in a class, 90 students pass in Accounts, 80 students pass in Economics, and 65 students pass in Business Studies. Out of these, 55 students passed in Accounts and Economics, 40 students passed in Economics and Business Studies, and 45 students passed in Accounts and Business Studies.

$$\begin{aligned} E_1 &= 3n - 45 \\ E_2 &= 140 - 3n \\ E_3 &= n \end{aligned}$$



$$\min n = 20$$

$$\max n = 40$$

$$\max(2n - 15) = 2 \times 40 - 15 = 65$$

$$\max(100 - 2n) = 100 - 2 \times 20 = 60$$

Questions



#Q. What is the minimum number of students who pass in at-least one subject? $95 + x = 95 + 20 = 115$

#Q. What is the maximum number of students who pass in atleast 2 subjects? $E_{x_2} + E_{x_3} = (140 - 3x) + x = 140 - 2x = 140 - 2 \times 20 = 100$

#Q. What is the maximum number of students who pass in Accounts or Economics but not in Business Studies? $x + 30 = 40 + 30 = 70$

#Q. What is the minimum number of students who pass in Economics only? $x - 15 = 20 - 15 = 5$

#Q. What is the minimum number of students who do not pass in Economics? 80

Among the 160 students in a class, 90 students pass in Accounts, 80 students pass in Economics, and 65 students pass in Business Studies. Out of these, 55 students passed in Accounts and Economics, 40 students passed in Economics and Business Studies, and 45 students passed in Accounts and Business Studies.



Thank You



THANK
You

